Exam

Electricity and Electronics EBN 122

12 November 2019

Information			
Maximum marks:	92	Full marks:	90
Duration of test:	180 min + (5 min before and 15 min after for OCR)	Open/Closed book:	Closed
Total number of pages (this page included)		36	

Important

1. The examination regulations of the University of Pretoria apply.
2. Answer all questions on the OCR forms provided. The question number **01** in this question paper refers to row 01 of the OCR form, etc.
3. Read the instructions on the OCR form carefully.
4. Where applicable, answers must include units.
5. Final answers must be written as real numbers and not as fractions.
6. Use at least 3 significant numbers to write irrational numbers.

Lecturers: Prof J.P. de Villiers, Prof J.P. Jacobs, Dr X. Ye

INSTRUCTIONS

Do step 0 in the first 5 minutes of the session. (-5 to 0 min)

Do step 1 in the next 180 minutes of the test. (0-180 min)

Do step 2 within the next 15 minutes. (180-195 min)

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form, and also fill in your name and student number on additional pages of the OCR form.

STEP 1 (0 min to 180 min): Do your own calculations on the question paper. Write down the answers for numeric questions on the PRACTICE ANSWER SHEET. Be sure to include the units where applicable.

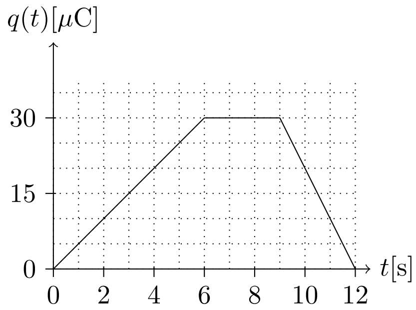
STEP 2 (180 min to 195 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2019EBN122E01**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

Section 1 / Afdeling 1

Question 1 [1]

The graph of the charge in an initially uncharged capacitor is shown in the figure below.

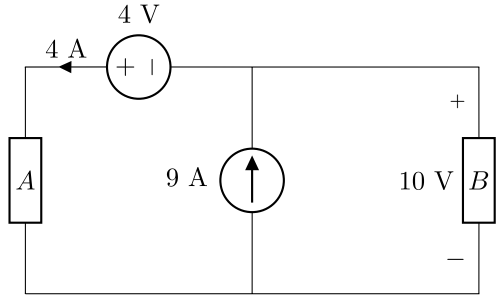


01	Find the current i_c at $t = 10$ s. [1]

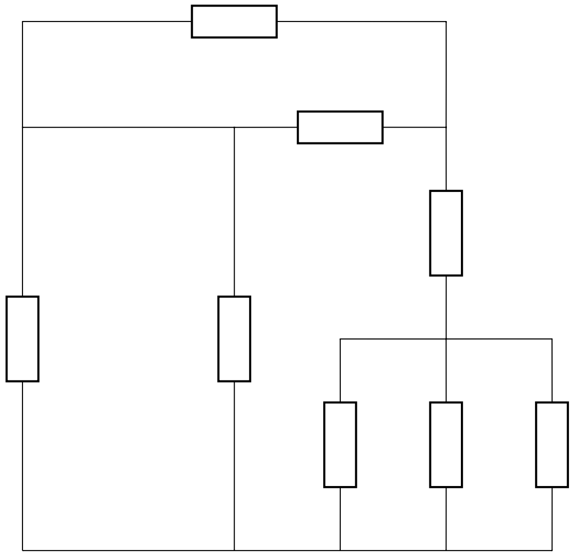
Question 2 [2]

The voltage across a 5 Ω resistor increases linearly from 0 mV to 15 mV between times $t = 0$s and $t = 20$s.

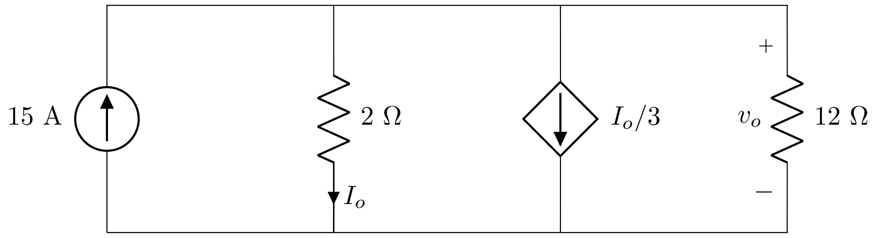
02	Calculate the charge Q_R that flows through the resistor between $t = 5$s and $t = 15$s. [2]

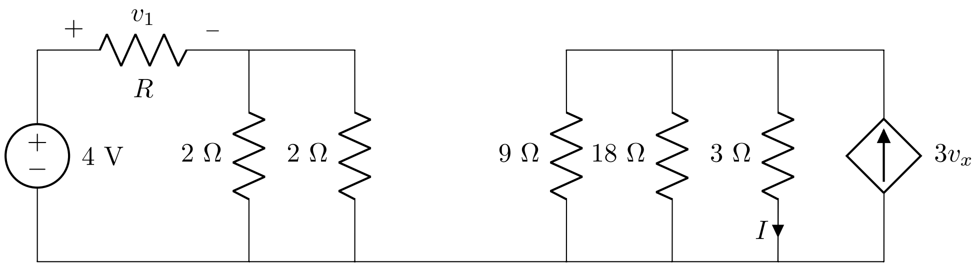
Questions 3 to 5 [3]	
Calculate the power relating to the following elements in the circuit below:	
	
03	Calculate the power P_A associated with element A. [1]
04	Is element A supplying power (write a 1) or absorbing power (write a 2)? [1]
05	Calculate the resistance R_B of circuit element B. [1]

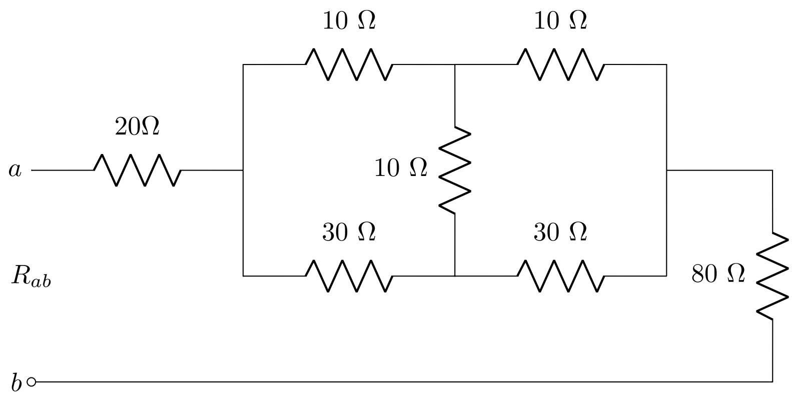
Questions 6 to 8 [3]



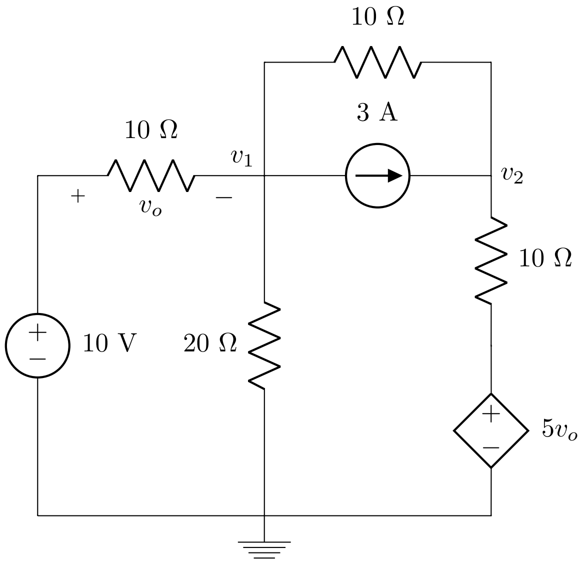
06	Determine the number of nodes n in the circuit above. [1]
07	Determine the number of branches b in the circuit above. [1]
08	Determine the number of independent loops l in the circuit above. [1]

Questions 9 to 10 [3]	
	
09	Determine v_o. [2]
10	Determine I_o [1]

Questions 11 to 12 [3]	
	
11	If $v_1 = 3$ V, determine R. [2]
12	If $v_x = 1.5$ V, determine I. [1]

Question 13 [2]	
	
13	Determine R_{ab}. [2]

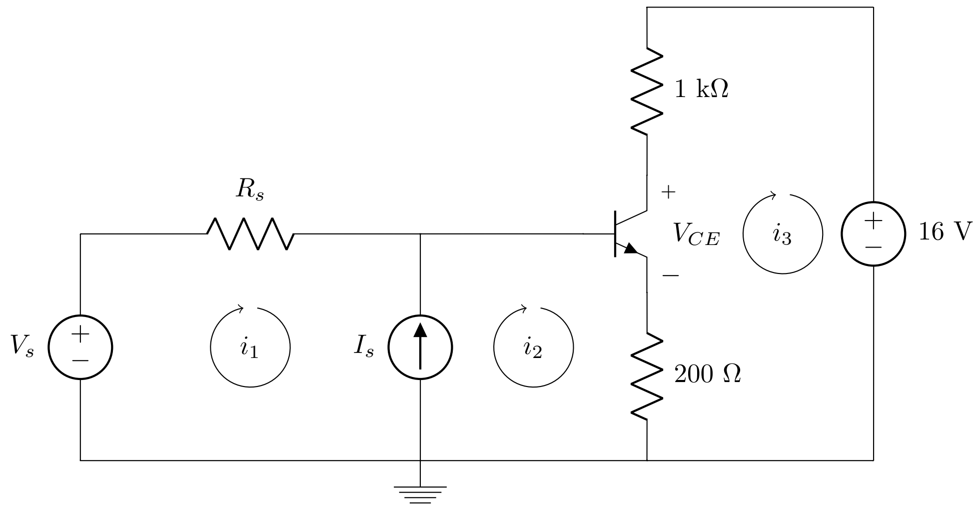
Questions 14 to 17 [4]
Use nodal analysis to set up the following two equations of the form $$av_1 - 2v_2 = b,$$ $$cv_1 + dv_2 = -50.$$ Calculate the unknown coefficients.



14	a [1]	**15**	b [1]	**16**	c [1]	**17**	d [1]

Questions 18 to 22 [5]

Consider the circuit below containing a transistor with $\beta = 99$ and $V_{BE} = 0.7$ V.



Assuming $V_s = 2$V, $R_s = 100$kΩ and $I_s = 3\mu$A, use mesh analysis to set up the following two equations of the form

$$ei_1 + fi_2 = 1.3,$$
$$gi_1 + hi_2 = 3.0 \times 10^{-6}.$$

Calculate the unknown coefficients e, f, g and h.

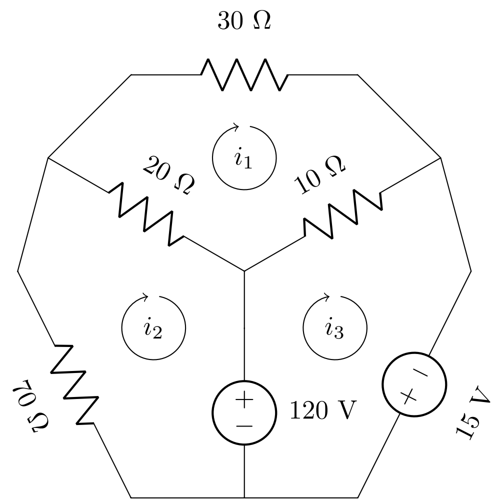
18	e [1]	**19**	f [1]	**20**	g [1]	**21**	h [1]
22	Determine the value of V_{CE} if $i_2 = 13.333$ μA. [1]						

Questions 23 to 26 [4]

Use inspection to compile a matrix equation of the form below

$$\begin{bmatrix} i & j & l \\ j & m & 0 \\ l & 0 & p \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -120 \\ r \end{bmatrix}$$

and provide the values of the matrix elements asked for below.



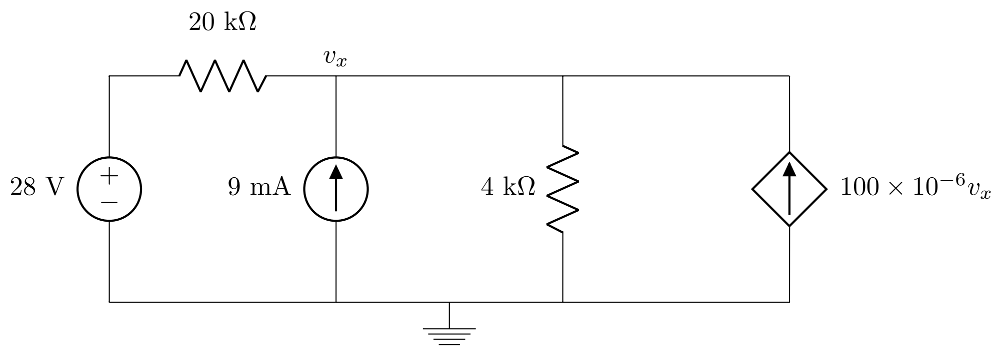
23	i [1]	**24**	m [1]	**25**	l [1]	**26**	r [1]

(Page for rough work)

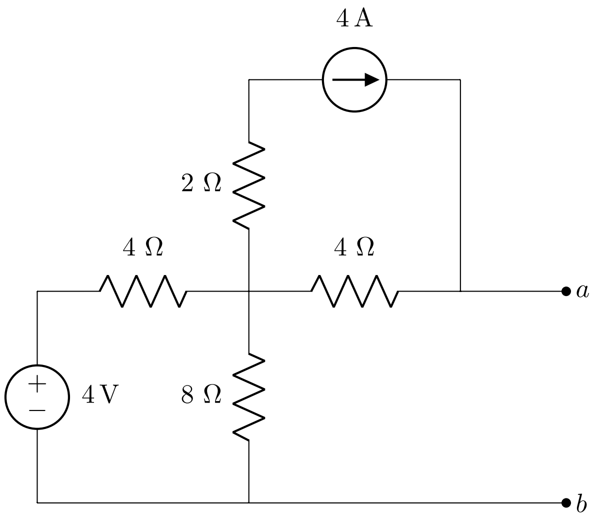
Total Section 1 / Totaal Afdeling 1: [30]

Section 2 / Afdeling 2

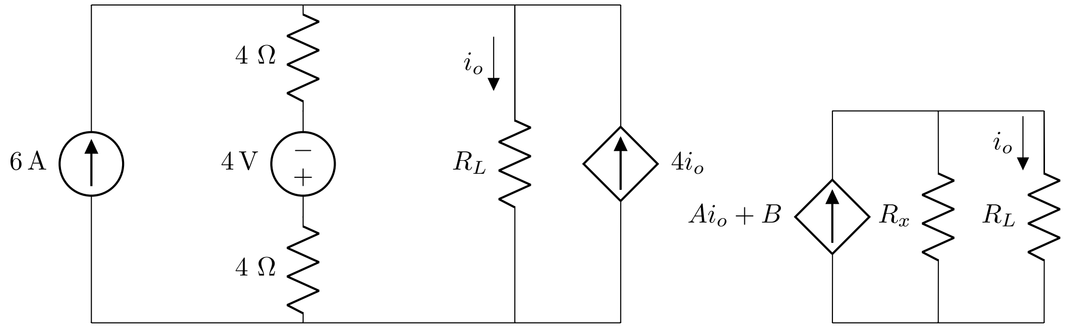
Questions 27 to 28 [4]



27	Determine the contribution to v_x of the 28 V voltage source. [2]
28	Determine the contribution to v_x of the 9 mA current source. [2]

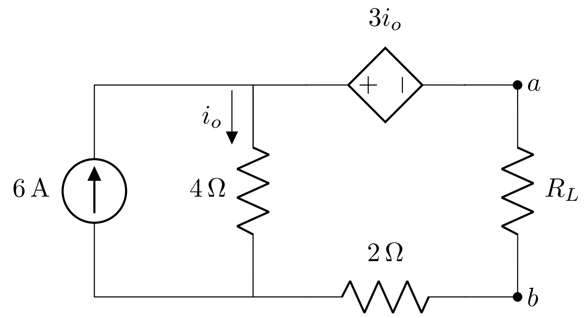
Questions 29 to 30 [4]	
Find the Norton equivalent current I_N, and Norton equivalent resistance R_N, with reference to terminals $a - b$.	
	
29	I_N [2]
30	R_N [2]

Questions 31 to 33 [4]	
Simplify the circuit on the left by means of source transformations so that it looks like the circuit on the right. Determine R_x, A and B.	



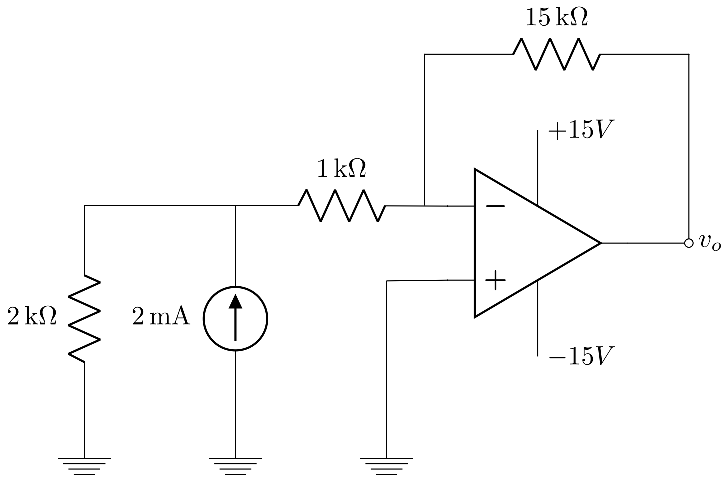
31	R_x [1]
32	A [1]
33	B [2]

Questions 34 to 35 [4]	
Find the Thevenin equivalent voltage V_{Th} seen by the load R_L, and the value of R_L so that maximum power is transferred to R_L.	

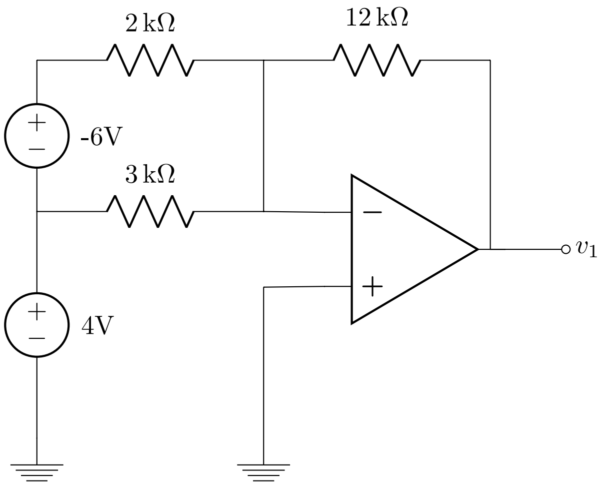


34	V_{Th} [2]
35	R_L [2]

Question 36 [2]
Assume that the op-amp is ideal. Find v_o.

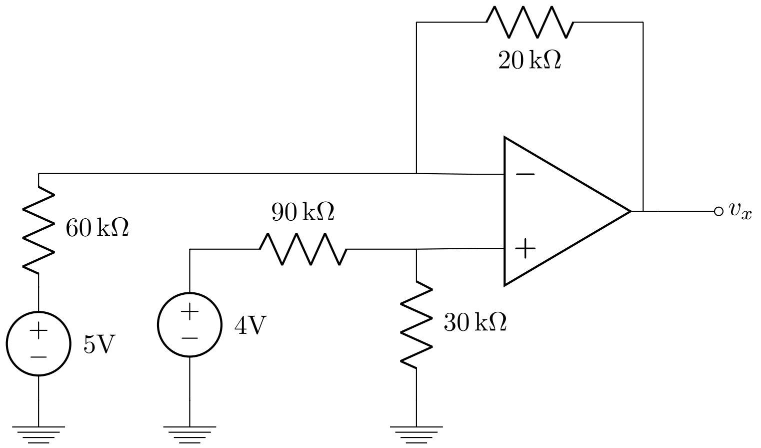


36	v_o [2]

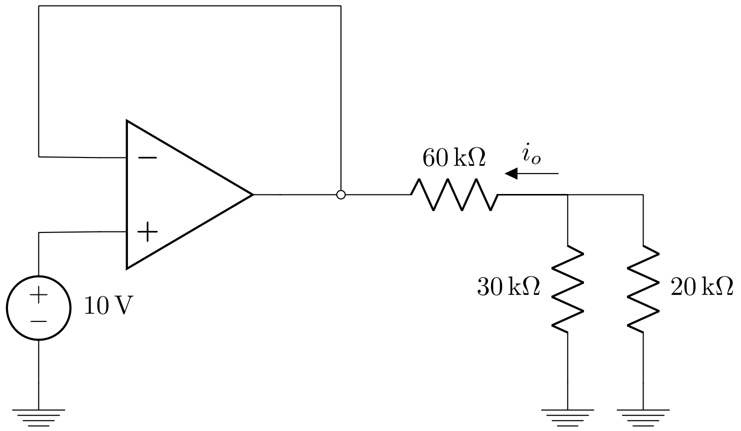
Question 37 [2]
Assume that the op-amp is ideal. Find v_1.


37	v_1 [2]

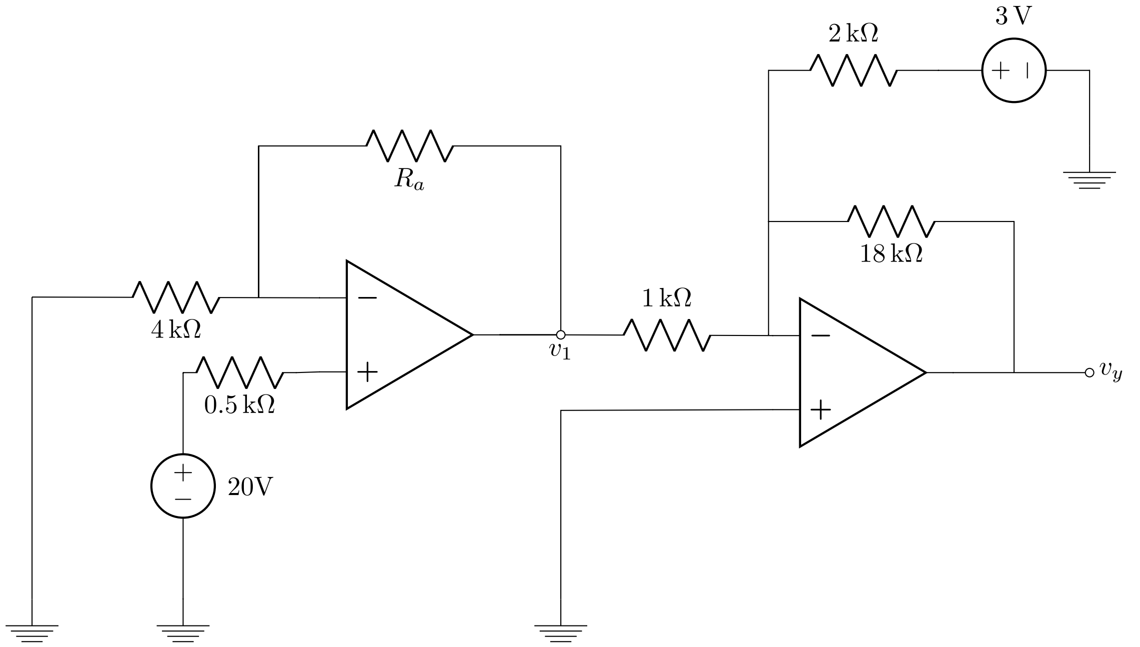
Question 38 [2]	
Assume that the op-amp is ideal. Find v_x.	

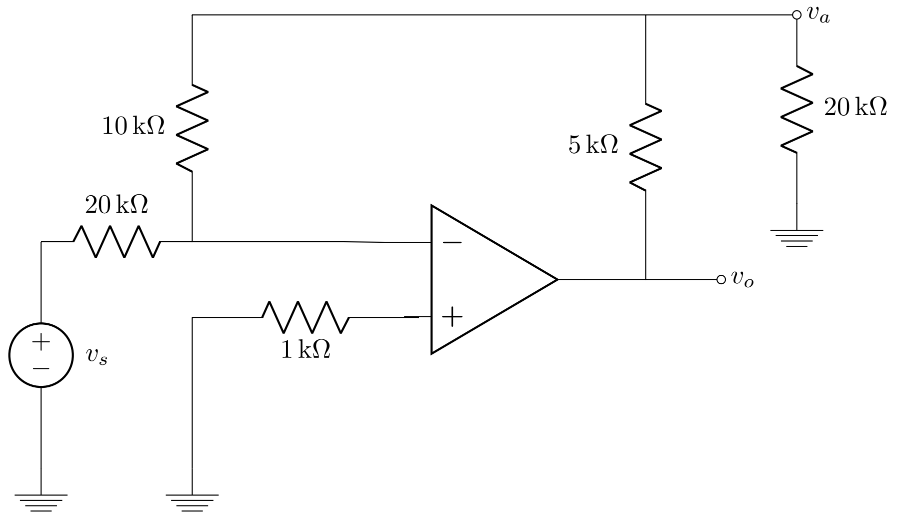


38	v_x [2]

Question 39 [2]
Assume that the op-amp is ideal. Find i_o.


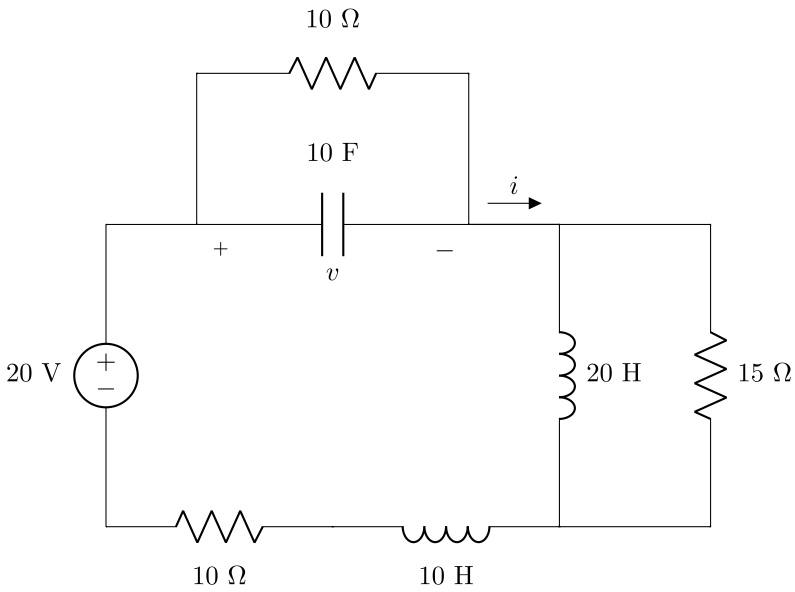
39	i_o [2]

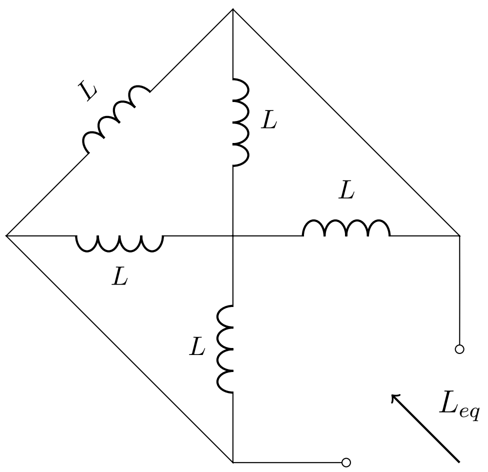
Questions 40 to 41 [3]	
Two ideal op-amps are connected as shown. It is known that v_1= 30 V. Find R_a and v_y.	
	
40	R_a [1]
41	v_y [2]

Questions 42 to 43 [4]	
Assume that the op-amp is ideal, and that v_a=10 V. Find v_s and v_o.	
	
42	v_s [2]
43	v_o [2]

Total Section 2 / Totaal Afdeling 2: [31]

Section 3

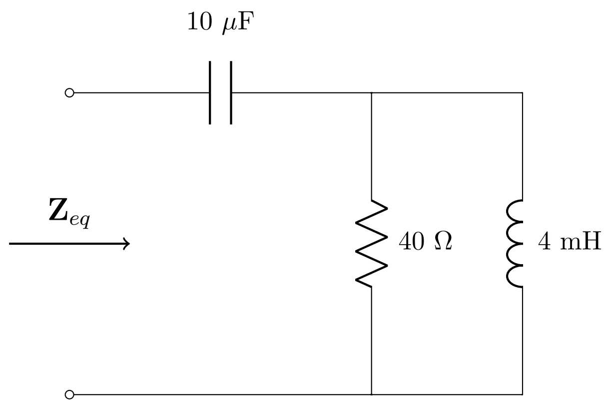
Questions 44 to 47 [4]	
The circuit below operates under steady-state dc conditions.	
	
44	Determine i. [1]
45	Determine v. [1]
46	Determine the energy E_L stored in the 10 H inductor. [1]
47	Determine the energy E_C stored in the capacitor. [1]

Question 48 [2]
Consider the circuit below.


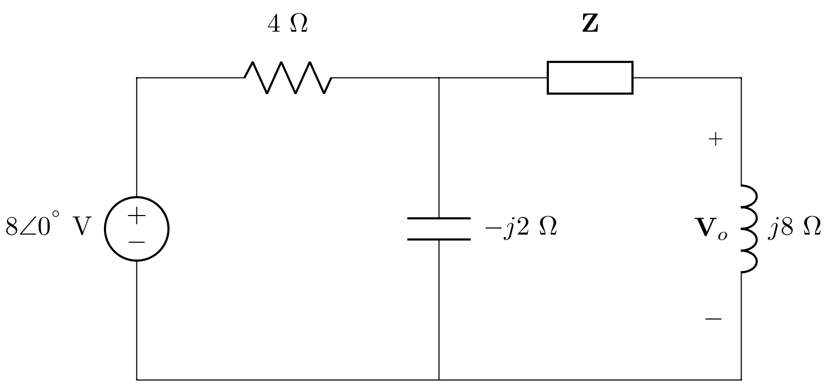
48	Find L_{eq} if $L = 10$ H. [2]

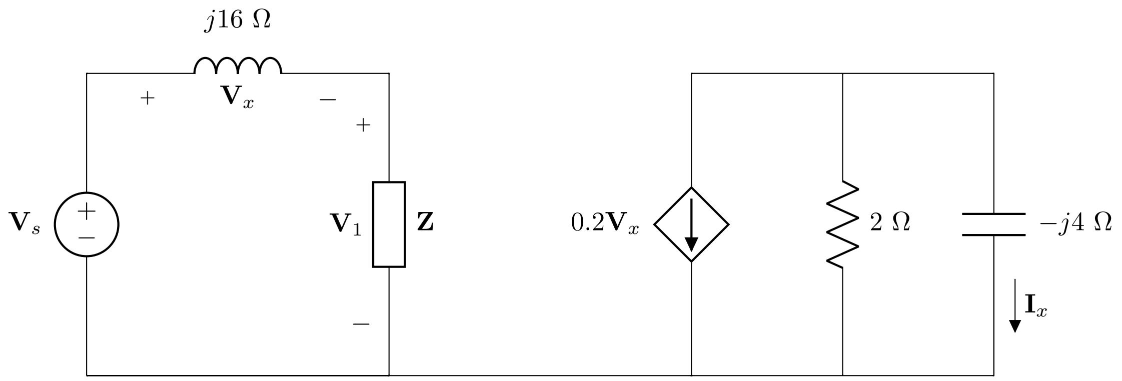
Questions 49 to 51 [4]	
Consider three sinusoid waveforms $v_1(t)$, $v_2(t)$, and $v_3(t)$, where $$v_1(t) = 5\sin\left(\omega t + 30^\circ\right), \text{ and}$$ $$v_2(t) = 12\cos\left(\omega t + 60^\circ\right).$$ The waveforms $v_1(t)$, $v_2(t)$, and $v_3(t)$ can be denoted by phasors $\mathbf{V}_1$, $\mathbf{V}_2$, and $\mathbf{V}_3$, respectively. The phase angle between $\mathbf{V}_2$ and $\mathbf{V}_3$ is ϕ_0, and the phase angle between $\mathbf{V}_1$ and $\mathbf{V}_3$ is $3\phi_0$; $\mathbf{V}_3$ is in the 1st quadrant of the phasor domain.	
49	Choose the correct statement. 1. v_2 leads v_3. 2. v_1 leads v_3. 3. v_1 leads v_2. Write 1 if **statement 1** is correct, 2 if **statement 2** is correct, **or** 3 if **statement 3** is correct. [2]
50	Find $\phi_0 \in [0^\circ, 180^\circ]$. [1]
51	Use phasors to find $\mathbf{V} = \mathbf{V}_1 - \mathbf{V}_2$ in **polar** form. [1]

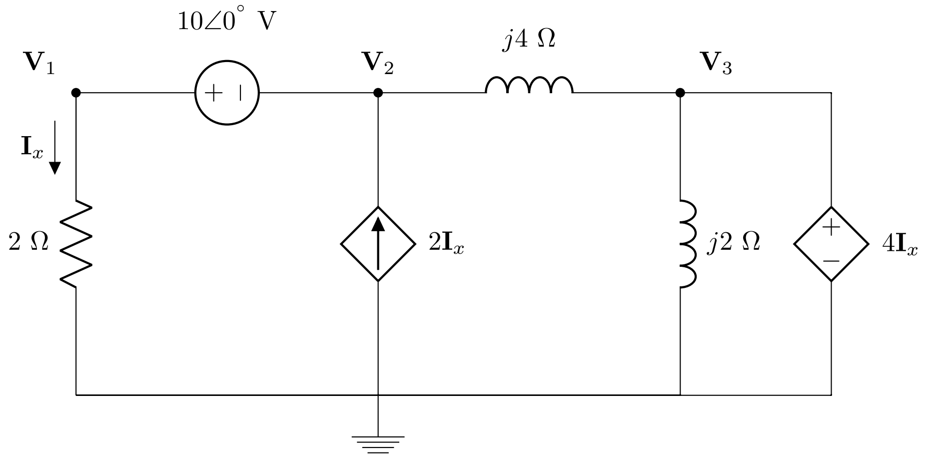
Question 52 [2]	
Consider the circuit below.	

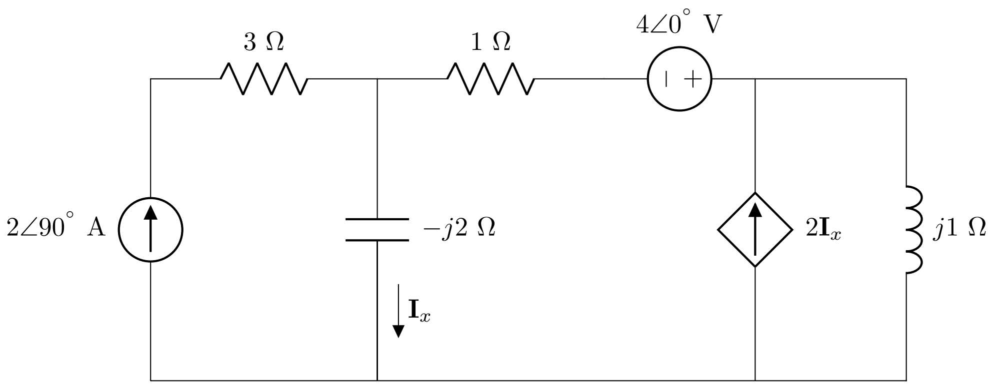


52	Find $\mathbf{Z}_{eq}$ in **rectangular** form at an angular frequency $\omega = 10^4$ rad/s. [2]

Question 53 [2]	
Consider the circuit below.	
	
53	Find **Z** when $\mathbf{V}_o = 16\angle 0^\circ$ in **rectangular** form. [2]

Questions 54 to 55 [4]	
Consider the circuit below.	
	
54	Given $\mathbf{Z} = 1 + j2\ \Omega$ and $\mathbf{V}_x = 8\angle 0^\circ$ V, find $\mathbf{V}_s$ in **rectangular** form. [2]
55	If $\mathbf{I}_x = 4\angle 0^\circ$ A, find $\mathbf{V}_x$ in **rectangular** form. [2]

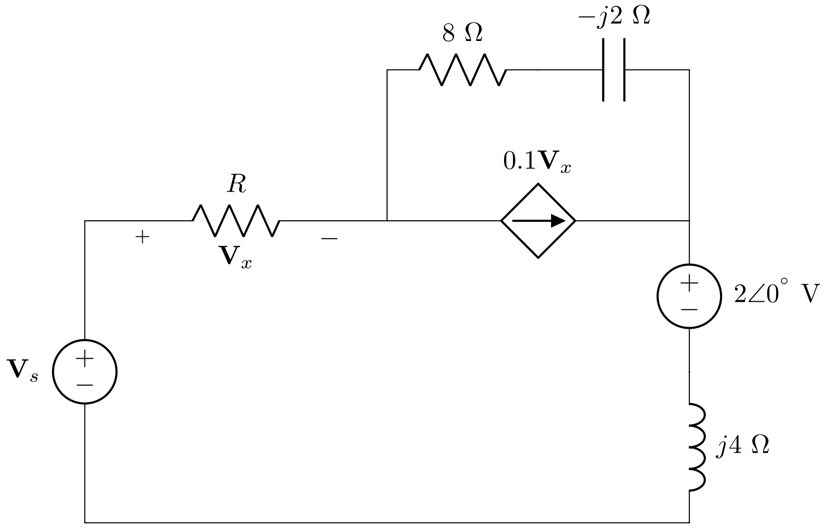
Questions 56 to 59 [4]	
Use nodal analysis to set up two equations of the following form: $$\mathbf{a}\mathbf{V}_1 + \mathbf{b}\mathbf{V}_2 = 10\angle 0^\circ,$$ $$\mathbf{c}\mathbf{V}_1 + \mathbf{V}_2 = \mathbf{d}.$$ Find the complex values $\mathbf{a}, \mathbf{b}, \mathbf{c}$, and $\mathbf{d}$ in **rectangular** form.	
	
56	Determine **a**. [1]
57	Determine **b**. [1]
58	Determine **c**. [1]
59	Determine **d**. [1]

Question 60 [2]	
Consider the circuit diagram below.	
	
60	Find the contribution to $\mathbf{I}_x$ from the voltage source in **rectangular** form. [2]

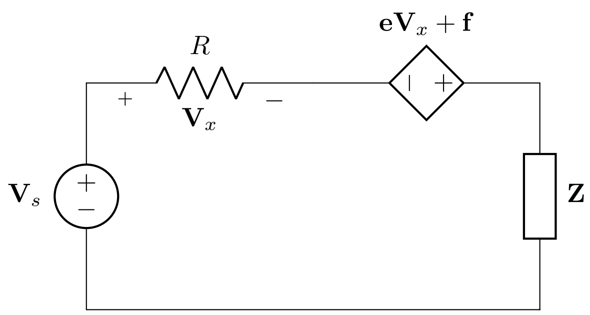
Questions 61 to 63 [3]

Apply source transformation to transform circuit ***A*** to circuit ***B***.

Circuit A:



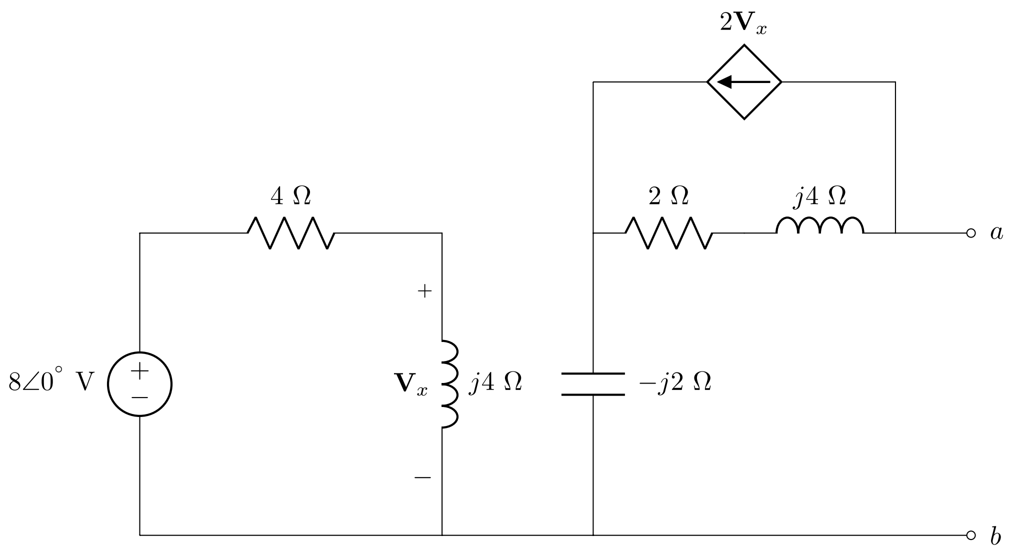
Circuit B:



61	Find $\mathbf{Z}$ in **rectangular** form. [1]
62	Find $\mathbf{e}$ in **rectangular** form. [1]
63	Find $\mathbf{f}$ in **rectangular** form. [1]

Turn over for additional space for Questions 61–63.

Page for rough work.

Questions 64 to 65 [4]	
Consider the circuit diagram below.	
	
Find the Thévenin equivalent voltage $\mathbf{V}_{Th}$ and impedance $\mathbf{Z}_{Th}$ at terminals $a-b$.	
64	Determine $\mathbf{V}_{Th}$ in **rectangular** form. [2]
65	Determine $\mathbf{Z}_{Th}$ in **rectangular** form. [2]

Total Section 3: [31]

Grand Total: [92]

PRACTICE ANSWER SHEET

Assessment ID	2	0	1	9	E	B	N	1	2	2	E	0	1

How to enter complex numbers:										
5	∠	-	9	0	°				V	
6	.	9	∠	1	5	.	9	°	μ	A
1	2	-	j	8						
j	1	5							Ω	
-	1	.	5	+	j	8	.	2	m	V

Rectangular:	j is at the start of the imaginary number.
	The j is in the CENTRE of the cell.
	The j does NOT cross or touch cell boundaries!
Polar:	The degree sign ° is directly after the angle value.

Please round all irrational numbers to a minimum of 3 significant digits.

	Answer	Unit
	SECTION 1	
01	$i_c =$	
02	$Q_R =$	
03	$P_A =$	
04	1 or 2	
05	$R_B =$	No unit required
06	$n =$	
07	$b =$	No unit required
08	$l =$	No unit required
09	$v_o =$	
10	$I_o =$	
11	$R =$	
12	$I =$	
13	$R_{ab} =$	
14	$a =$	No unit required
15	$b =$	No unit required
16	$c =$	No unit required
17	$d =$	No unit required

18	$e =$	No unit required
19	$f =$	No unit required
20	$g =$	No unit required
21	$h =$	No unit required
22	$V_{CE} =$	
23	$i =$	No unit required
24	$m =$	No unit required
25	$l =$	No unit required
26	$r =$	No unit required
	SECTION 2	
27	Contribution of 28 V source:	
28	Contribution of 9 mA source:	
29	$I_N =$	
30	$R_N =$	
31	$R_x =$	
32	$A =$	No unit required
33	$B =$	No unit required
34	$V_{Th} =$	
35	$R_L =$	
36	$v_o =$	
37	$v_1 =$	
38	$v_x =$	
39	$i_o =$	
40	$R_a =$	
41	$v_y =$	
42	$v_s =$	
43	$v_o =$	

SECTION 3		
44	$i =$	
45	$v =$	
46	$E_L =$	
47	$E_C =$	
48	$L_{eq} =$	
49	Write **1**, **2**, or **3** in OCR form.	No unit required.
50	$\phi_0 =$	
51	$\mathbf{V} =$	No unit required.
52	$\mathbf{Z_{eq}} =$	
53	$\mathbf{Z} =$	
54	$\mathbf{V}_s =$	
55	$\mathbf{V}_x =$	
56	$a =$	No unit required.
57	$b =$	No unit required.
58	$c =$	No unit required.
59	$d =$	No unit required.
60	Contribution to $\mathbf{I}_x$ from voltage source =	
61	$\mathbf{Z} =$	
62	$e =$	No unit required.
63	$f =$	No unit required.
64	$\mathbf{V}_{Th} =$	
65	$\mathbf{Z}_{Th} =$	